

Field Trial of Failure Localization in a Backbone Optical Network

Camille Delezoide⁽¹⁾, Petros Ramantanis⁽¹⁾, Lluís Gifre⁽¹⁾, Fabien Boitier⁽¹⁾, and Patricia Layec⁽¹⁾

⁽¹⁾ Nokia Bell Labs France, camille.delezoide@nokia-bell-labs.com

Abstract We propose a robust and easy-to-deploy failure localization method trialed on a backbone network carrying live traffic to generate strategic network health insights, notably regarding Q-drops.

Introduction

The automated localization of optical layer failures is a strategic step in many applications aimed at optimizing network reliability and reducing OPEX, such as proactive maintenance. Historically, localization solutions focused on leveraging extra hardware dedicated to active or passive monitoring^{[1]–[8]}. Lately, solutions relying on the aggregation of readily available monitoring data from typical network equipment, e.g. coherent transponders, tend to gain momentum^{[9]–[14]}. These solutions fully leverage the existing infrastructure, thus do not involve significant CAPEX. Their common drawback however is the ambiguity arising when one set of observations has multiple explanations.

In this context, we propose an algorithm leveraging monitoring data to first list all possible failure scenarios, then to determine the most likely failure localization. We evaluated it during a 10-month field trial in a transparent optical backbone network carrying live traffic. To do so, we installed coherent transponders at two distant locations to build a total of four distinct lightpaths delivering each 100Gb/s with PDM-QPSK modulation format. We polled the values of pre and post-FEC BER for the four lightpaths every second. We first describe the localization algorithm with a focus on ambiguous cases. Then, we report the observation of Q-drops during the field trial, underlining the need to automate the localization of the origins of post-FEC errors. We finally present and discuss the localization results obtained during the field trial.

Localization Algorithm

In any transparent optical network, a soft or hard failure such as a fiber damage necessarily impacts all lightpaths going through the failed equipment. Thus, by monitoring for instance the quality of transmission (QoT) of all lightpaths at reception, it is possible to narrow the search for the failure cause, and eventually localize it depending

on the topology. This is essentially what our algorithm does. For compacity, we will describe the various stages of the algorithm depicted in Fig. 1 through the example of the field trial topology, focusing on use cases to outline the various localization challenges and implemented solutions.

Alarm generation and aggregation: Our localization algorithm relies on the routing matrix $\mathbf{R}$ - mapping lightpaths over segments - and the alarm vector $\mathbf{A}$ aggregating similar alarms from the multiple monitored lightpaths. Depending on the application, the alarms can be derived from any monitored metrics. In the field trial, we focused on issues having a direct impact on the quality of service by generating alarms upon the detection of post-FEC errors. The main challenge of the alarm aggregation is to handle unsynchronized data flows which, as we will see later, can lead to localization errors. A simple solution is to introduce a correlation window starting when a first lightpath alarm rises (cf. Fig. 2). Then, all alarms rising within the correlation window are assumed to be part of the same event, as opposed to distinct events.

Segment sorting: Segments here are groups of links interconnecting two nodes. The first stage of our algorithm sorts segments according to their potential involvement in a studied alarm. This is achieved by making boolean comparisons between the alarm vector $\mathbf{A}$ and all columns of the routing matrix $\mathbf{R}$ representing the various segments. This can either be made live each time $\mathbf{A}$ changes, or pre-emptively for all possible alarm vectors once the routing matrix is known. Let us consider the alarm $\mathbf{A}_1 = (1 \ 0 \ 0 \ 1)$ for our field trial topology, generated when post-FEC errors are detected for both LP₁ and LP₄ within the same correlation window. From the routing matrix, we identify five scenarios leading to $\mathbf{A}_1$ and involving three segments, S₁, S₂ and S₇.

First, $\mathbf{A}_1$ could be generated upon a failure of S₁. This is the most likely scenario considering that it involves a single segment failure, as op-

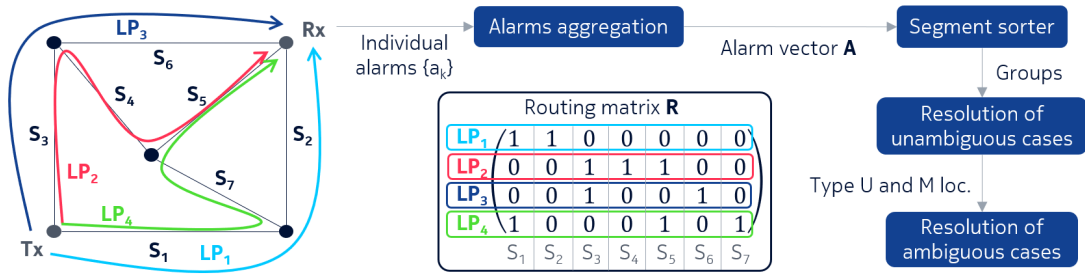


Fig. 1: Field trial topology (not to scale), associated routing matrix, and stages of the failure localization algorithm

posed to multiple failures in the other scenarios. Indeed, the same alarm could be due to simultaneous failures of both S_2 and S_7 . It is also possible that A_1 is due to simultaneous failures on both S_1 and S_2 , or on S_1 and S_7 , or on S_1 , S_2 and S_7 .

Segments whose single failures fully explain an alarm such as S_1 for A_1 form *group U* (for unique). In contrast, segments whose failures need to be combined to fully explain an alarm - such as S_2 and S_7 here - form *group M* (for multiple).

Resolution of unambiguous cases: This stage consists in checking the ambiguity of the alarm and eventually solving unambiguous cases. Depending on how groups U and M are populated, there are only two possible types of unambiguous cases directly leading to the localization of the faulty segment(s). The first - referred as Type U - occurs when group U contains a single segment and group M is empty. Here, the fault can only come from the sole group U segment. This is the case for the alarm $(0\ 0\ 1\ 0)$ which can only come from a failure of S_6 . The second - referred as Type M - occurs when group U is empty and only one combination of group M segments fully explains the alarm. This is the case for the alarm $(0\ 0\ 1\ 1)$ which can only be generated upon failures of both S_6 and S_7 . In all other cases, the alarm is ambiguous and requires a final stage to be solved.

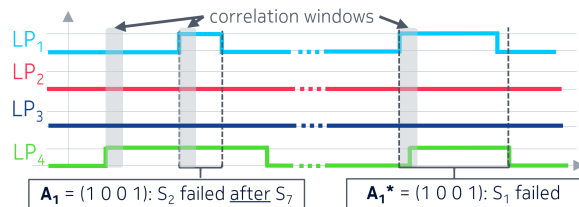


Fig. 2: Two cases of disambiguation from alarm history of the same field trial alarm $A_1=(1\ 0\ 0\ 1)$.

Resolution of ambiguous cases: Up to this point, the algorithm acted in a memoryless fashion, i.e. only processing the current alarm status. We now leverage the chronology of alarms to solve ambiguous cases, as depicted in Fig. 2. In our field trial topology, we show two cases where $A_1=(1\ 0\ 0\ 1)$ is generated. In the first case, A

goes from $(0\ 0\ 0\ 1)$ to $(1\ 0\ 0\ 1)$. While $(0\ 0\ 0\ 1)$ directly points to a S_7 failure, $(1\ 0\ 0\ 1)$ is ambiguous. Yet, the most likely explanation for the $(1\ 0\ 0\ 1)$ alarm is that S_2 failed after S_7 . In the second case, because both LP_1 and LP_4 alarms rise within the correlation window, we assume that A directly goes from $(0\ 0\ 0\ 0)$ to $(1\ 0\ 0\ 1)$. Here, although it is theoretically possible that both S_2 and S_7 failed within the correlation window, the most likely explanation is that S_1 failed. Such process can be used recursively in any network to find the most likely localization of any ambiguous alarm. The clear limitation of failure localization through the aggregation of end receiver-based alarms is that for ambiguous cases, the most likely solution is not necessarily the correct one. The smaller the correlation delay, the less this should occur. In practice, the correlation delay will be limited by the sampling frequency of the alarm status, and/or by the maximal amount of desynchronization between any two alarm streams. Optimization aside, assumed or deduced segment failure statistics allow the estimation and comparison of the likelihoods of all possible failure scenarios generating the same alarms. In turn, this information can be used to quantify the uncertainty on any network health metric deriving from the localization, such as the segment availability.

Q-drops as localized, self-repaired outages

In addition to the major outages and maintenance windows reported during the trial, the analysis of the complete dataset revealed a significant number of short-spanned events - typically less than a minute - where the reported post-FEC BER was non-zero for a single or multiple lightpaths at the same time. We found these observations to be consistent with the Q-drops described in^{[15],[16]} and^[17], an important difference being that Q-drops were initially defined as outlier Q^2 values above the FEC limit, thus assumed fundamentally different from regular outages impacting the quality of service. Yet, the collected Q^2 values used in these former studies, i.e. averages

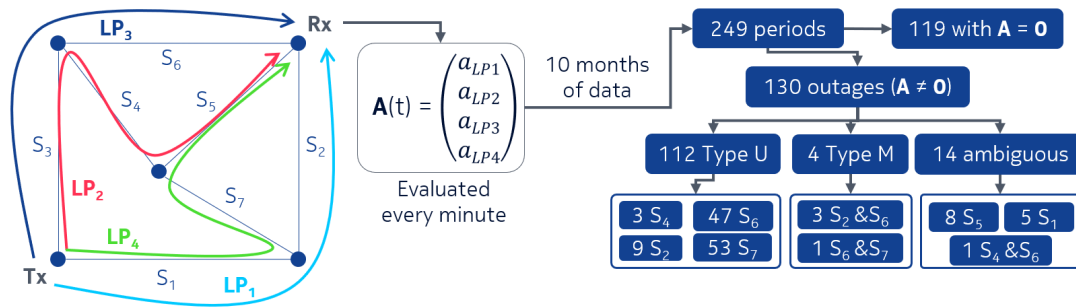


Fig. 3: Summary of the localization results for the full field trial

over 15-minute-windows, did not allow to determine how low the instantaneous Q^2 fell during a Q-drop. Thanks to the one-second-resolution we achieved in this field trial, we verified as illustrated in Fig. 4 that all detectable Q-drops were actually short, self-repaired bursts of post-FEC errors of unknown origin. A direct application of the localization algorithm we applied to the dataset has been to pinpoint which segments were particularly affected, as a first step towards the mitigation of this service-impacting phenomenon.

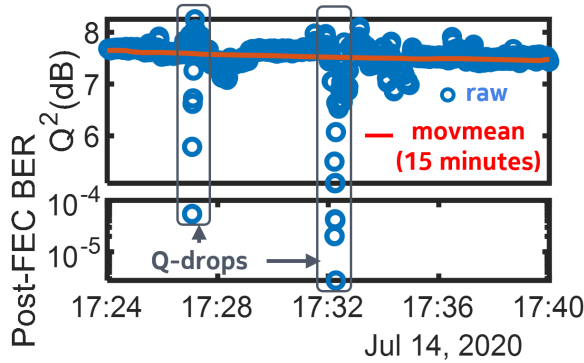


Fig. 4: Example of Q-drops observed during our field trial.

Localization results for the field trial

We applied our localization algorithm to identify, among the seven segments traversed by the four monitored lightpaths, which segment(s) was(were) involved in the presence of post-FEC errors, i.e. Q-drops and regular outages. We evaluated $\mathbf{A}$ at each minute of the trial, setting $a_j=1$ when at least one post-FEC BER sample for LP_j was non-zero, and $a_j=0$ otherwise. We then regrouped consecutive identical alarms to form 249 time periods during which $\mathbf{A}$ remained constant. Out of these 249 periods, 130 had a non-zero alarm status. After solving unambiguous cases, the exact origin of the alarm for most - i.e. 116 - of these periods were known, and 14 periods with ambiguous alarms remained. The high ratio of unambiguous alarms is due to the relative simplicity of the field trial topology. More complex topologies should result in higher ratios of ambiguous

alarms, especially if the lightpath diversity is low compared to the number of segments.

The last stage of our algorithm, leveraging alarm status history with a one-minute correlation delay finally provided the most likely localizations for the 14 ambiguous cases left to solve. The process is recapitulated in Fig. 3. Overall, 97% of periods with non-zero alarms were due to single segment failures, while 3% were due to double failures. These ratios are expected considering that the probability of a double failure, the product of two single failure probabilities, is generally speaking much lower than a typical single failure probability. From the failure counts, we observed that S_6 was involved in 39% of all outaged periods, and 42% for S_7 . This can be surprising considering that S_6 is actually 16 times longer than S_7 . The analysis of the extents of the outaged periods involving S_6 and S_7 showed that both had similar ratios of total segment downtime divided by segment length. We were thus able to conclude that the abnormal rate of failures localized to S_7 was explained by a much higher frequency of Q-drop events compared to other segments. A fact we were then able to verify from the dataset.

Conclusion

We proposed an algorithm to localize the origin of alarms generated upon the analysis of monitoring data collected from coherent transponders in transparent optical networks. We applied this algorithm in a field trial to study occurrences of post-FEC errors in a backbone network carrying live traffic, and demonstrated how the localization can be leveraged to study and improve the reliability of optical networks.

Acknowledgments

The authors wish to warmly thank Sylvain Chenard for fruitful discussions, and our silent partners for granting us access to their network, and for their multiple contributions that made this study possible.

References

- [1] I. Sankawa, S. Furukawa, Y. Koyamada, and H. Izumita, "Fault location technique for in-service branched optical fiber networks", *IEEE Photonics Technology Letters*, vol. 2, no. 10, pp. 766–768, 1990. DOI: 10.1109/68.60785.
- [2] C. Mas, I. Tomkos, and O. Tonguz, "Failure location algorithm for transparent optical networks", *IEEE Journal on Selected Areas in Communications*, vol. 23, no. 8, pp. 1508–1519, 2005. DOI: 10.1109/JSAC.2005.852182.
- [3] Yonggang Wen, V. Chan, and Lizhong Zheng, "Efficient fault-diagnosis algorithms for all-optical WDM networks with probabilistic link failures", *Journal of Lightwave Technology*, vol. 23, no. 10, pp. 3358–3371, 2005. DOI: 10.1109/JLT.2005.855695. (visited on 05/04/2021).
- [4] S. S. Ahuja, S. Ramasubramanian, and M. Krunz, "SRLG Failure Localization in All-Optical Networks Using Monitoring Cycles and Paths", in *IEEE INFOCOM*, 2008, p. 9.
- [5] J. Tapolcai, B. Wu, and P.-H. Ho, "On Monitoring and Failure Localization in Mesh All-Optical Networks", in *IEEE INFOCOM 2009 - The 28th Conference on Computer Communications*, Rio De Janeiro, Brazil, Apr. 2009, pp. 1008–1016. DOI: 10.1109/INFCOM.2009.5062012.
- [6] S. Stanic and S. Subramaniam, "Fault Localization in All-Optical Networks with User and Supervisory Light-paths", in *ICC proceedings*, Kyoto, Japan: IEEE, 2011, pp. 1–6. DOI: 10.1109/icc.2011.5962755.
- [7] Y. Xuan, Y. Shen, N. P. Nguyen, and M. T. Thai, "Efficient Multi-Link Failure Localization Schemes in All-Optical Networks", *IEEE Transactions on Communications*, vol. 61, no. 3, pp. 1144–1151, 2013. DOI: 10.1109/TCOMM.2013.012313.110645.
- [8] A. P. Vela, B. Shariati, M. Ruiz, *et al.*, "Soft Failure Localization During Commissioning Testing and Light-path Operation", *Journal of Optical Communications and Networking*, vol. 10, no. 1, A27, 2018. DOI: 10.1364/JOCN.10.000A27.
- [9] K. Christodoulopoulos, N. Sambo, and E. Varvarigos, "Exploiting Network Kriging for Fault Localization", in *Optical Fiber Communication Conference*, OSA, 2016, W1B.5. DOI: 10.1364/OFC.2016.W1B.5.
- [10] C. Delezoide, K. Christodoulopoulos, A. Kretsis, *et al.*, "Pre-Emptive Detection and Localization of Failures Towards Marginless Operations of Optical Networks", in *ICTON proceedings*, 2018. DOI: 10.1109/ICTON.2018.8473928.
- [11] T. Panayiotou, S. P. Chatzis, and G. Ellinas, "Leveraging Statistical Machine Learning to Address Failure Localization in Optical Networks", *Journal of Optical Communications and Networking*, vol. 10, no. 3, p. 162, 2018. DOI: 10.1364/JOCN.10.000162.
- [12] S. Barzegar, E. Virgillito, M. Ruiz, *et al.*, "Soft-Failure Localization and Device Working Parameters Estimation in Disaggregated Scenarios", in *Optical Fiber Communication Conference (OFC) 2020*, OSA, 2020, Th1F.2. DOI: 10.1364/OFC.2020.Th1F.2.
- [13] Z. Li, Y. Zhao, Y. Li, S. Rahman, X. Yu, and J. Zhang, "Demonstration of Fault Localization in Optical Networks Based on Knowledge Graph and Graph Neural Network", in *Optical Fiber Communication Conference (OFC) 2020*, San Diego, California: OSA, 2020, Th1F.5. DOI: 10.1364/OFC.2020.Th1F.5.
- [14] R. Liang, F. Liu, and J. Liu, "A Belief Network Reasoning Framework for Fault Localization in Communication Networks", *Sensors*, vol. 20, no. 23, p. 6950, Dec. 2020. DOI: 10.3390/s20236950.
- [15] M. Ghobadi and R. Mahajan, "Optical Layer Failures in a Large Backbone", in *IMC proceedings*, ACM Press, 2016, pp. 461–467. DOI: 10.1145/2987443.2987483.
- [16] D. W. Boertjes, M. Reimer, and D. Côté, "Practical considerations for near-zero margin network design and deployment (invited)", *J. Opt. Commun. Netw.*, vol. 11, no. 9, pp. C25–C34, 2019. DOI: 10.1364/JOCN.11.000C25.
- [17] C. Delezoide, P. Ramantanis, and P. Layec, "Leveraging field data for the joint optimization of capacity and availability in low-margin optical networks", *Journal of Lightwave Technology*, vol. 38, no. 24, pp. 6709–6718, 2020. DOI: 10.1109/JLT.2020.3022107.